

SSC CGL Fractions Unit Digit Shortcut Sheet

SSC CGL / Tier Prep / Quantitative Aptitude

Premium Snapshot

Exam: SSC CGL

Subject: Quantitative Aptitude

Topic: Fractions Unit Digit Digital Sum

This Study Hub premium pack is a guided sheet. Keep official syllabus, official notices, and standard books as the source of truth.

Study Flow

- Step 1: Open the matching syllabus or textbook folder first.
- Step 2: Convert this sheet into a one-page note and a short practice set.
- Step 3: Solve PYQ, mock, or board-style questions from the same folder family.
- Step 4: Add every mistake to one error log and revise it weekly.

Premium Content

- Percentage to fraction recall makes profit-loss, SI-CI, ratio, mixture, DI, and discount questions much faster.
- Use unit digit and last two digit checks before long multiplication when options are available.
- Use digital sum as a filter, not as a proof, because different numbers can share the same digital sum.

Percentage to Fraction Chart

- $1/2=5\%$ | $1/3=33.33\%$ | $1/4=25\%$ | $1/5=20\%$ | $1/6=16.67\%$
- $1/7=14.29\%$ | $1/8=12.5\%$ | $1/9=11.11\%$ | $1/10=10\%$ | $1/11=9.09\%$
- $1/12=8.33\%$ | $1/13=7.69\%$ | $1/14=7.14\%$ | $1/15=6.67\%$ | $1/16=6.25\%$
- $1/17=5.88\%$ | $1/18=5.56\%$ | $1/19=5.26\%$ | $1/20=5\%$ | $1/21=4.76\%$
- $1/22=4.55\%$ | $1/23=4.35\%$ | $1/24=4.17\%$ | $1/25=4\%$ | $1/26=3.85\%$
- $1/27=3.7\%$ | $1/28=3.57\%$ | $1/29=3.45\%$ | $1/30=3.33\%$

Speed Rules

- Base 50 square: $(50+x)^2 = (25+x) \times x^2$. Example: $53^2 = 28|09 = 2809$.
- Base 50 square: $(50-x)^2 = (25-x) \times x^2$. Example: $47^2 = 22|09 = 2209$.
- Carry rule: if x^2 has three digits, carry the hundreds to the left part. Example: $67^2 = 42|289 = 4489$.
- Base 100 square: $97^2 = (100-2x) \times x^2$ where $x=3$, so $94|09 = 9409$.
- Last two digit pattern: square endings repeat through complements, so learn 1 to 25 endings strongly.
- Unit digit rule: even number cube stays even, odd number cube stays odd; use last digit to remove wrong options.
- Unit digit square: last digit 0,1,4,5,6,9 only. If option ends in 2,3,7,8, reject it for a square.
- Digital sum: reduce calculation and option to one digit by repeatedly adding digits; use it as a fast option filter.
- Divisibility by 9: digit sum divisible by 9 means number is divisible by 9.
- Divisibility by 11: difference between alternate digit sums divisible by 11 means number is divisible by 11.
- Pi questions: when $22/7$ appears, check options with 7, 11, 22, or 154 factor before doing full multiplication.

Output Checklist

- One syllabus map or chapter map is complete.
- One practice set is attempted in timed mode.
- Wrong answers are grouped by concept, careless error, or time pressure.
- Next revision date is written before closing this pack.